

Vaggelis (Evangelos) Dorovatas

vdorovatas@hotmail.gr | ed3061@nyu.edu | +30 6988702353

Google Scholar | GitHub | LinkedIn

Research Interests

- Representation Learning, Neural Network Architectures / Connectivity
- Memory, Continual Learning, In-Context Learning, Long Context Processing

Education

New York University, PhD in Data Science, advisor: Prof. Pratyusha Sharma	Sept 2026 – Jun 2031
National Technical University of Athens, MEng in Electrical & Computer Eng.	Sept 2019 – Jul 2025

Experience

Toyota Motor Europe – Zaventem, Belgium	Sept 2025 – Aug 2026
---	----------------------

- Role: AI Researcher
- Advisor: Dr. Rahaf Aljundi
- Topic: Memory, Continual Learning at Scale
- Outcome: **ICML '26 spotlight position** [2], CVPR '26 [9], TMLR '26 [6], Continual Learning from experience [4 - Under Review].

Archimedes Research Center – Athens, Greece	Mar 2025 – Aug 2025
---	---------------------

- Role: AI Research Intern
- Advisor: Prof. Alexandros Potamianos
- Topic: NN Architectures, Inter-Layer Connectivity & Information Flow, Data-efficient Deep Learning
- Outcome: **Awarded a European grant (EuroHPC)** to support our research on connectivity; **EMNLP '25 workshop winner and oral** [5].

Toyota Motor Europe – Zaventem, Belgium	Sept 2024 – Mar 2025
---	----------------------

- Role: AI Research Intern
- Advisor: Dr. Rahaf Aljundi
- Topic: Memory design for efficient streaming long video understanding with video-language models.
- Also involved in research on Personalization of large vision-language models [6].
- Outcome: **NeurIPS '25** publication [3].

NTUA/SLP lab – Athens, Greece	Jul 2023 – Aug 2024
-------------------------------	---------------------

- Role: AI student Researcher
- Advisor: Prof. Alexandros Potamianos
- Topic: Connectivity Patterns and Information Flow in Neural Networks
- Outcome: *Auto-Compressing Networks* [1], **NeurIPS '25 oral** (top 0.36%).

Prom Racing Student Team NTUA – Athens, Greece	Oct 2023 – Aug 2024
--	---------------------

- Role: Autonomous Driving Engineer

Agile Actors – Athens, Greece	Summer 2022
-------------------------------	-------------

- Role: Software Engineering Intern (DevOps)

Awards/Honors

- European grant (EuroHPC) to support our research on NN connectivity and incorporate auto-compressing and other hybrid connectivity architectures into large language models.
- EMNLP 2025 BabyLM Workshop (Accelerating language modeling with cognitively plausible datasets) strict track winner award.
- Invited oral presentation, NeurIPS 2025.

Publications

1. *Auto-Compressing Networks (NeurIPS '25 oral)* Paper: [arXiv](#)
Vaggelis Dorovatas, Georgios Paraskevopoulos, Alexandros Potamianos
2. *Position: Modular Memory is the Key to Continual Learning Agents (ICML '26 spotlight position)* Paper: [arXiv](#)
Vaggelis Dorovatas et al.
3. *Recurrent Attention-based Token Selection for Efficient Streaming Video-LLMs (NeurIPS '25)* Paper: [arXiv](#)
Vaggelis Dorovatas, Soroush Seifi, Gunshi Gupta, Rahaf Aljundi
4. *Continual Self-Improvement with Lightweight Experiential Latent Memories (Under Review for NeurIPS '26)* Preprint: [arXiv](#)
Vaggelis Dorovatas, Nancy Kalaj, Rahaf Aljundi
5. *Masked Diffusion Language Models with Frequency-Informed Training (EMNLP '25 workshop oral)* Paper: [arXiv](#)
Despoina Kosmopoulou, Efthymios Georgiou, **Vaggelis Dorovatas**, Georgios Paraskevopoulos, Alexandros Potamianos
6. *Personalization Toolkit: Training Free Personalization of Large Vision Language Models (TMLR'26)* Preprint: [arXiv](#)
Soroush Seifi, **Vaggelis Dorovatas**, Daniel Olmeda Reino, Rahaf Aljundi
7. *Online In-Context Distillation for Low-Resource Vision Language Models* Preprint: [arXiv](#)
Zhiqi Kang, Rahaf Aljundi, **Vaggelis Dorovatas**, Karteek Alahari
8. *Hybrid ACNs: Unifying Auto-Compressing and Residual Architectures* Preprint: [link](#)
Vaggelis Dorovatas, Alexandros Potamianos
9. *Ego: Embedding-Guided Personalization of Vision-Language Models (CVPR '26)* Paper: [arXiv](#)
Soroush Seifi, Simon Gardier, **Vaggelis Dorovatas**, Daniel Olmeda Reino, Rahaf Aljundi